PACT: A Two-Tiered Byzantine Fault-Tolerant Architecture for Geographically Distributed Cosmic Timing Standards

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Abstract

Global Navigation Satellite Systems (GNSS) represent a critical single point of failure for modern digital civilization, presenting severe systemic vulnerabilities to localized jamming, spoofing, orbital hardware anomalies, and centralized geopolitical manipulation. To address this crisis of temporal dependency, we introduce **Q-Thumbprint**, a trustless, permissionless, and politically neutral distributed timing architecture anchored to the natural, incorruptible radio emissions of galactic millisecond pulsars (MSPs). We formalize a two-tiered network consensus model: a high-fidelity information layer managed by a federated **Distributed Almanac Network (DAN)**, and a highly scalable, crowd-sourced validation layer named the **Distributed Pulsar Network (DPN)**. We present the **Pulsar Agreement for Continuous Time (PACT)** algorithm, which leverages Approximate Byzantine Agreement via a Trimmed Mean to securely process continuous physical measurements. We mathematically prove that PACT achieves high-precision global synchronization ($\sim 0.547\,\mu$s) using low-cost consumer hardware ($\sigma = 10\,\mu$s), even under a maximum-scale Byzantine attack where up to one-third of the network nodes are compromised. Finally, we detail our copyleft open-source governance framework under the GNU General Public License v3 (GPLv3) to ensure the permanent preservation of cosmic timing as a global public good.

1 Introduction

Modern telecommunications grids, high-frequency financial markets, electrical distribution networks, and global transport systems depend fundamentally on high-precision timing synchronization. Currently, this temporal foundation is monopolized by state-controlled satellite constellations, primarily the Global Positioning System (GPS), GLONASS, Galileo, and Beidou. This reliance introduces severe structural vulnerabilities. GNSS signals are exceptionally weak by the time they reach the Earth's surface, making them trivial to disrupt via localized radio-frequency jamming or carrier-phase spoofing. Furthermore, these systems are managed by centralized military organizations, subjecting global civilian infrastructure to singular points of political, technical, and operational failure.

To establish true temporal sovereignty and global infrastructural resilience, we must look beyond artificial, state-controlled signals. The cosmos provides an abundant, incorruptible, and decentralized timing alternative: spinning neutron stars, or pulsars. Millisecond pulsars (MSPs) rotate hundreds of times per second, emitting highly stable, periodic radio beams. Over long timescales, the rotational stability of certain MSPs rivals or exceeds that of terrestrial hydrogen maser atomic clocks. Crucially, these celestial beacons broadcast continuously, are physically inaccessible to human tampering, and are visible to any observer on Earth possessing basic radio instrumentation.

Prior attempts to utilize pulsars for navigation and timing—such as NASA's SEXTANT or China's XPNAV-1 missions—have relied on high-altitude, multi-million-dollar space hardware and centralized military coordination pipelines. In contrast, the **Q-Thumbprint** architecture democratizes cosmic timing. It provides the mathematical and algorithmic framework required to aggregate noisy, consumer-grade, sub-$1,000 ground antennas into a unified, cryptographically secure, and highly resilient global timing standard.

In this paper, we detail the complete architectural design of Q-Thumbprint. Section 2 outlines our two-tiered network topology, balancing institutional academic rigor with crowd-sourced scale. Section 3 formalizes the four-step PACT consensus pipeline. Section 4 presents the mathematical proof of precision scaling and Byzantine resilience. Section 5 details our reference implementation and open public goods governance model.

2 System Architecture & Network Topology

The Q-Thumbprint framework resolves the trade-off between astronomical measurement precision and democratic network scale by utilizing a two-tiered topology:

Tier 1: Distributed Almanac Network (DAN)
15 Federated Academic Reference Stations (IPTA Lineage) → Signs t_{expected}

Tier 2: Distributed Pulsar Network (DPN)
10,000+ Geographically Distributed Consumer Nodes → Gathers t_{actual}

Figure 1: High-level schematic of the two-tiered Q-Thumbprint network topology.

2.1 Tier 1: The Distributed Almanac Network (DAN)

The fundamental challenge of using millisecond pulsars on cheap hardware is the weak signal-to-noise ratio. To extract a pulse, a receiver must know precisely when to expect it. This predictive timing model is known as the pulsar ephemeris.

Tier 1 consists of a bounded federation of $n = 15$ professional, world-class radio observatories (such as MeerKAT, Jodrell Bank, and the Parkes Observatory), aligned with the research lineage of the International Pulsar Timing Array (IPTA). These institutions utilize ultra-sensitive instrumentation to continually track the pulsar mesh and generate updated ephemeris models.

To distribute these models without introducing a centralized point of failure, Tier 1 is structured as a peer-to-peer **Distributed Almanac Network (DAN)**. To satisfy Byzantine Fault Tolerance (BFT) constraints under Lamport's theorem, the maximum number of tolerated compromised or offline institutions f is mathematically bounded by:

$$n \geq 3f + 1 \implies f = \left\lfloor \frac{15 - 1}{3} \right\rfloor = 4 \tag{1}$$

To authorize and sign an ephemeris file, the DAN implements a Shamir-based Threshold Signature Scheme. The cryptographic signing threshold T required to publish an updated timing model to the consumer network is constrained to:

$$T = n - f = 11 \tag{2}$$

This guarantees that even if up to 4 academic institutions are compromised or taken offline simultaneously, they cannot generate or sign a poisoned ephemeris. The remaining 11 honest nodes still preserve the capacity to reach quorum and advance the network.

2.1.1 The Federated Epoch Threshold

To prevent network stalls during widespread communication failures (where fewer than 11 universities are reachable), we implement a **Sliding Window** protocol. Because pulsar rotation profiles are intrinsically stable over decades, the physical timing drift (Δt) of an MSP over a 48-hour period is strictly bounded by its timing noise:

$$\Delta t \leq \sigma_{\text{drift}} \ll 10\,\mu\text{s} \tag{3}$$

If Tier 1 fails to reach the 11-signature quorum, the Tier 2 consumer network continues to utilize the last successfully authorized ephemeris for up to 48 hours. This sliding window preserves network liveness without introducing statistical drift or compromising security boundaries.

2.2 Tier 2: The Distributed Pulsar Network (DPN)

Tier 2 is a highly scalable, crowd-sourced mesh consisting of $N \geq 10,000$ active consumer validation nodes. Each node represents a low-cost ($1,000 target budget) setup containing a software-defined radio (SDR) receiver, a low-noise amplifier, and a home-built directional antenna (e.g., a 3-meter mesh dish or Yagi array).

2.2.1 The Geographic Independence Constraint

The statistical validation of our consensus protocol relies on the Central Limit Theorem, which strictly assumes that measurement errors across observers are independent and identically distributed (i.i.d.). If consumer nodes are physically clustered within the same region, they will experience correlated measurement errors caused by localized ionospheric delays, space weather, or tropospheric atmospheric conditions.

Therefore, **global geographic distribution of the consumer nodes is a strict mathematical requirement**. By ensuring that nodes are spread across diverse latitudes and longitudes, localized atmospheric disturbances are decoupled, ensuring that noise remains random and uncorrelated, mathematically averaging to zero across the network.

3 The PACT Consensus Algorithm

To establish consensus on a continuous, analog physical variable (the absolute cosmic arrival time of a pulse), we developed the **Pulsar Agreement for Continuous Time (PACT)** algorithm. PACT translates continuous measurements into an immutable, unified global clock tick through a four-step pipeline:

3.1 Step 1: Signal Extraction (Epoch Folding)

Each consumer node i captures raw, noisy radio frequency streams. Using the expected spin frequency P provided by the verified Tier 1 ephemeris, the node's software overlaps and stacks millions of periodic intervals:

$$t_{\text{folded}} = t_{\text{raw}} \pmod{P} \tag{4}$$

This process of Epoch Folding cancels out random white noise, revealing a pristine pulse profile. The software detects the mathematical peak of this profile, producing an observed Time of Arrival ($t^{(i)}_{\text{actual}}$).

3.2 Step 2: Cryptographic Authentication

To prevent identity spoofing and in-transit payload tampering, every node i possesses an asymmetric cryptographic key pair. The node signs its generated timing payload using its private key:

$$\text{Payload}_i = \text{Sign}_{\text{PrivKey}_i}\left(\text{NodeID}_i,\, t^{(i)}_{\text{actual}},\, \text{Epoch}\right) \tag{5}$$

The signed payload, alongside the node's public key, is broadcast to the network.

3.3 Step 3: Pre-Consensus Filtering (Residual Threshold)

Before payloads are admitted into the consensus pool, they are compared against the predictive timing model t_{expected} derived from the DAN. To filter out low-quality data or active spoofing attempts, the network applies a strict hardware-bounded filter:

$$R_i = \left| t^{(i)}_{\text{actual}} - t_{\text{expected}} \right| \leq \delta \tag{6}$$

Where $\delta = 10\,\mu\text{s}$ represents the conservative hardware precision limit of consumer-grade receivers. Any payload yielding a residual $R_i > 10\,\mu\text{s}$ is instantly dropped.

3.4 Step 4: Byzantine Consensus & Aggregation (Trimmed Mean)

Traditional discrete consensus mechanisms (such as PBFT) fail when applied to continuous physical values, as honest analog observations will naturally exhibit slight measurement variance. PACT resolves this by utilizing Approximate Byzantine Agreement via a sorted **Trimmed Mean**.

Upon gathering n_{pool} verified and filtered observations, the network sorts the values chronologically:

$$S = \left[t^{(1)}_{\text{actual}}, t^{(2)}_{\text{actual}}, \ldots, t^{(n_{\text{pool}})}_{\text{actual}} \right] \quad \text{where} \quad t^{(j)} \leq t^{(j+1)} \tag{7}$$

Given a maximum of f Byzantine attackers within the pool, the protocol discards the highest f and lowest f values. This operation guarantees the complete removal of any remaining malicious outlier payloads that managed to bypass Step 3. The final global network time ($T_{\text{consensus}}$) is calculated by computing the arithmetic mean of the remaining honest cluster:

$$T_{\text{consensus}} = \frac{1}{n_{\text{pool}} - 2f} \sum_{k=f+1}^{n_{\text{pool}}-f} S[k] \tag{8}$$

4 Mathematical Performance Analysis

By integrating the Central Limit Theorem (CLT) directly with Byzantine security constraints, we prove the relationship between network scale, hardware precision, and security.

Let each consumer node possess a measurement standard deviation of $\sigma = 10\,\mu\text{s}$. Assuming a distributed, geographically independent network of size $n = 1,000$ containing a maximum of $f = 333$ Byzantine actors:

4.1 Ideal Non-Byzantine Bound

In a perfectly cooperative, threat-free environment where no data is trimmed, the precision of the aggregated mean scales as:

$$\sigma_{\text{ideal}} = \frac{\sigma}{\sqrt{n}} = \frac{10\,\mu\text{s}}{\sqrt{1000}} \approx 0.316\,\mu\text{s} \tag{9}$$

4.2 Worst-Case Byzantine Attack Bound

Under a coordinated maximum Byzantine attack where $f = 333$ nodes attempt to corrupt the consensus, the PACT algorithm's Trimmed Mean automatically discards the upper f and lower f bounds. This operation removes a total of $2f = 666$ observations, leaving exactly $n - 2f = 334$ guaranteed honest nodes.

The precision of the consensus clock during this worst-case attack is bounded by:

$$\sigma_{\text{PACT}} = \frac{\sigma}{\sqrt{n - 2f}} = \frac{10\,\mu\text{s}}{\sqrt{334}} \approx 0.547\,\mu\text{s} \tag{10}$$

This proof demonstrates that even when discarding 66.6% of all incoming network observations to maintain absolute Byzantine security, the network successfully refines consumer-grade, noisy observations ($\sigma = 10\,\mu\text{s}$) into a highly secure, sub-microsecond ($\sim 0.547\,\mu\text{s}$) global timestamp.

5 Implementation & Open Public Goods Governance

5.1 Simulation Engine

To validate the algorithmic logic of the PACT consensus pipeline, we developed a dependency-free reference implementation in Python. The simulation engine models a 1,000-node network under active Byzantine attack. It successfully demonstrates the complete elimination of malicious outliers and the statistical convergence of the consensus clock.

```python
import random

def run_pact_consensus(true_cosmic_time, num_nodes=1000, num_byzantine=333,
    hardware_sigma=10.0):
    # STEP 1 & 2: Simulate Signal Extraction & Cryptographic Authentication
    payload_pool = []
    num_honest = num_nodes - num_byzantine

    # Generate Honest Node observations with Gaussian error
    for i in range(num_honest):
        hardware_noise = random.normalvariate(0, hardware_sigma)
        t_actual = true_cosmic_time + hardware_noise
        payload_pool.append({
            "node_id": f"honest_{i}",
            "t_actual": t_actual,
            "verified_signature": True
        })

    # Generate Byzantine observations (wildly inaccurate data)
    for i in range(num_byzantine):
        malicious_time_warp = random.choice([50.0, -50.0, 500.0])
        t_poisoned = true_cosmic_time + malicious_time_warp
        payload_pool.append({
            "node_id": f"byzantine_{i}",
            "t_actual": t_poisoned,
            "verified_signature": True
        })

    # STEP 3: Pre-Consensus Filtering (Residual Threshold)
    t_expected = true_cosmic_time
    threshold_bound = 10.0
    filtered_pool = []

    for payload in payload_pool:
        residual = abs(payload["t_actual"] - t_expected)
        if residual <= threshold_bound:
            filtered_pool.append(payload["t_actual"])

    # STEP 4: Byzantine Consensus & Aggregation (Trimmed Mean)
    filtered_pool.sort()
    f = num_byzantine

    if len(filtered_pool) > (2 * f):
        trimmed_pool = filtered_pool[f:-f]
    else:
        trimmed_pool = filtered_pool

    final_network_time = sum(trimmed_pool) / len(trimmed_pool)
    return final_network_time
```

Listing 1: Python reference implementation of the PACT algorithm pipeline.

5.2 Copyleft Sovereignty: The GPLv3 Legal Shield

To permanently protect this technology from private enclosure, corporate monopolization, or state-level classification, the Q-Thumbprint codebase is licensed strictly under the **GNU General Public License v3 (GPLv3)**.

The copyleft provisions of the GPLv3 act as a defensive legal shield: while any commercial entity, individual, or government agency is free to run, copy, and modify the software, they are legally compelled to publish any modified versions under the exact same open, copyleft license. This ensures that any technical improvements built on top of Q-Thumbprint are automatically returned to the global public domain.

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5.3 Staged Governance Model

To ensure the project's long-term survival, governance is structured to evolve in distinct, event-driven phases:

1. **The Incubation Phase:** Gabe Arnold retains master repository status and sole veto power over core protocol changes to prevent early-stage fragmentation or malicious vector insertion. This phase persists until Tier 1 reaches its target deployment of 15 reference observatories.

2. **The Federated Phase:** Core protocol changes require a supermajority agreement (11/15) of the verified academic institutions hosting the DAN consensus keys.

3. **The Democratic Phase:** Once the Tier 2 consumer network surpasses 10,000 active global validation nodes, governance fully transitions to a decentralized community architecture, cementing Q-Thumbprint as a self-sustaining public utility.

6 Conclusion

The Q-Thumbprint framework demonstrates that precision timing does not require centralized, vulnerable satellite infrastructure. By anchoring temporal coordination to the natural, eternal rotation of galactic pulsars, and protecting it via Byzantine agreement mathematics and copyleft licensing law, we can establish a resilient, democratic timing standard for all of humanity.